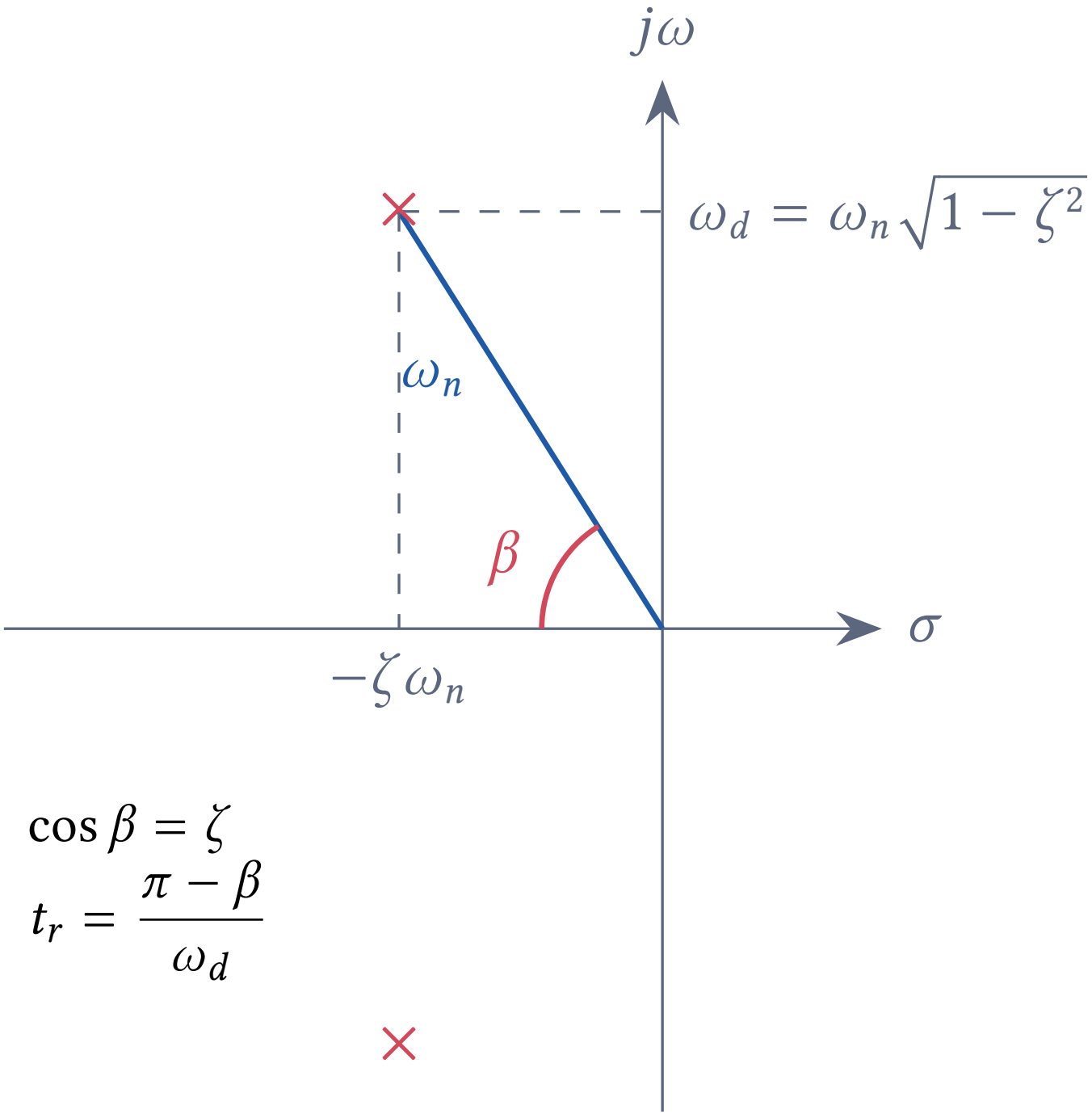


$$\cos\beta = \zeta$$

$$t_r = \frac{\pi - \beta}{\omega_d}$$